EVOLVING TECHNOLOGIES And TEST DEVICES For The Best 5G Experience

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Test methods for 5G ensure that the technology and services are implemented according to specifications, and quality of service matches the requirements of the application or service being delivered

Development in technology and network concepts for 5G requires test methods and processes to evolve. Test methods for 5G ensure that the technology and services are implemented according to specifications, and quality of service matches the requirements of the application or service being delivered. 5G chipsets validate the performance of 5G New Radio (NR) smartphones and Internet of Things (IoT) devices in the lab. These do not require access to expensive and complex 5G base stations.

Required features

Testing a fully data-centric 5G network with a diverse set of applications requires massive efforts in standalone testing. Analysis of the performance of such a network requires test automation, monitoring and built-in test systems. Interconnecting radio access elements with ultra-dense networks with backhaul architecture (using cloud networks) provides the ability to develop cloud-based test services. This allows testing of everything from anywhere.

Portable devices with Internet connectivity can be used by a remote user to analyse the systems and suggest changes to the field staff immediately. Handheld testing devices eliminate the need for multiple steps and create automated results.

Sometimes, results are also integrated to the server/cloud, so these can be monitored from the field remotely by user(s). In certain cases, built-in test features are inserted along with signals. For example, radio signal strengths can be analysed by taking information from social media channels.

Latest technologies

Madhukar Tripathi, head - marcom and optical business, Anritsu India Pvt Ltd, says, "MT8821C radio communication analyser is designed for R&D of mobile devices/user equipment such as smartphones, tablets and IoT modules."

Hari Kamalapuram, deputy general manager - communications business unit, Cyient Ltd, says, "5G/4G technologies require multiple testing areas including drive test and optimisation, KPI analysis, benchmark, broadband, protocol, and spectrum analysis for noise and electromagnetic environment.

"Earlier, equipment used for these tests comprised complex modular designs. Test results were processed in multiple steps. We have developed many tools for our internal consumption in planning and design.

"A handheld tool, i-netracker, is used for testing signal quality, call drops, throughput, career aggregation and so on for 4G/LTE networks. This test is popularly called single-cell functional test or single-cell verification test. The solution, being Android based, can be loaded into any Android-based cellphone. All measurement parameters are captured and sent to the central cloud. After completion, data can be processed remotely, and instant reports taken at any time."

There are two methods for measuring radiation patterns. One is using a large anechoic chamber or an over-the-air (OTA) system, which is expensive. Second is making something from scratch.

Phase-array antenna, beamforming algorithm and MMIC designers all need nearly continuous access for microwave (μWave) designs. The problem with the second option is that it is hard to do it right the first time. Many aspects of the problem are overlooked, including wire tangling, open software con-

Fig. 1: New-generation designs for 5G (Credit: Keysight)